```
        }
    }

    let message = "Hello"
    print(message.reversed())
```

Generics:

Generics allow you to write flexible and reusable code that can work with any type.

```
    func swapTwoValues<T>(_ a: inout T, _ b:
inout T) {
        let temporaryA = a
        a = b
        b = temporaryA
    }

    var first = "Hello"
    var second = "World"
    swapTwoValues(&first, &second)
    print("First: \(first), Second: \(second)")
```

SwiftUI:

SwiftUI is a modern framework for building user interfaces across all Apple platforms using Swift. It provides a declarative syntax for creating UI components.

```
    import SwiftUI

    struct ContentView: View {
        @State private var name = ""
        @State private var age = ""

        var body: some View {
            VStack {
                TextField("Enter your name",
text: $name)
                    .padding()
                    .textFieldStyle(RoundedBo
rderTextFieldStyle())

                TextField("Enter your age",
text: $age)
                    .padding()
                    .textFieldStyle(RoundedBo
rderTextFieldStyle())
                    .keyboardType(.numberPad)

                Button(action: {
                    let currentYear = Calendar.
current.component(.year, from: Date())
                    let birthYear = currentYear
- (Int(age) ?? 0)
                    print("Hello, \(name)! You
were born in \(birthYear).")
                }) {
                    Text("Greet")
                        .padding()
                        .background(Color.blue)
                        .foregroundColor(.white)
                        .cornerRadius(10)
                }
            }
            .padding()
        }
    }

    struct ContentView_Previews: PreviewPro-
vider {
        static var previews: some View {
            ContentView()
        }
    }
```

Industry Trends and Applications

Swift continues to gain popularity in various industries due to its performance and ease of use. Here are some current trends and applications of Swift:

iOS and macOS Development:

Swift is the primary language for developing applications for iOS and macOS. Its integration with Xcode and Cocoa/Cocoa Touch frameworks makes it the best choice for Apple platform development.

Server-Side Swift:

With the introduction of frameworks like Vapor and Kitura, Swift is being used for server-side development. This allows developers to write both client-side and server-side code in the same language, streamlining the development process.

Cross-Platform Development:

Swift's growth into server-side development and the introduction of Swift for TensorFlow (for machine learning) are expanding its use cases beyond just Apple platforms.

Machine Learning:

Swift for TensorFlow is an open-source project that brings Swift's modern language features to the TensorFlow machine learning framework, making it easier to write and maintain machine learning models.

Education:

Swift Playgrounds is an iPad app that makes learning Swift interactive and fun. It is widely used in educational settings to teach programming concepts to beginners.